

Echoes of Ideology: Evaluating Political Bias and Stability in Large Language Models Through Quantitative Analysis and Standardized Typology Testing

Abstract

This quantitative comparative analysis focuses on determining AI political bias and consistency. This research asks: “To what extent do GPT-4o, Claude 3.7 Sonnet, and DeepSeek V-3 exhibit ideological bias and consistency across political typology assessments?” Key gaps in the previous literature include the lack of inclusion of the new DeepSeek-V3 model, the absence of measures for LLM consistency in responses, and the lack of comparison between political typology tests for a given large language model (LLM). The research gathered data from Political Compass and 8Values, then standardized the 8Values data to be comparable to that of Political Compass. The data was then subjected to descriptive statistical analysis (mean, median, range, standard deviation, etc.) and inferential statistical analysis (one-way ANOVA test and Tukey HSD test) to determine whether the results differed significantly. ChatGPT demonstrated the highest level of consistency, while Claude was the most inconsistent. The 8Values test yielded less ideological extremity than the Political Compass test. All models leaned left, both socially (libertarian) and economically. Additionally, DeepSeek was statistically significantly more economically right-leaning than Claude ($p = 0.0459$ under a classical one-way ANOVA, though this result did not survive assumption-free robustness checks described below). There was no observed response-variability effect within a sitting, and all models had indistinguishable social and political ideologies (all p -values > 0.05).

This study was subsequently extended to twelve models across nine organizations (1,440 total administrations, a twelvefold increase), collected via an automated pipeline that scores both instruments against authoritative sources rather than a hand-derived conversion formula. Every model leaned in the same direction on every axis, replicating the original finding at substantially larger scale, and all omnibus tests (classical ANOVA, Welch’s ANOVA, Kruskal-Wallis) agreed at $p < .0001$ with large effect sizes ($\eta^2 = 0.575$ to 0.816) on every axis, a materially more robust result than the original three-model finding. The magnitude of lean varies substantially by model, and the original finding that Claude was the least consistent model does not generalize once more than one Claude generation is included: Claude Sonnet 4.5 was among the most consistent models tested and Claude Haiku 4.5 among the least.

1 Introduction

This study considers the cognitive implications of AI beyond automation. Technological advancements are often viewed as tools that automate labor, improving daily life efficiency. Such an assumption is based on historical precedent. The rise of inventions such as the steam engine, which mechanized power for factories, trains, and ships, revolutionized transportation and production.

Similarly, the invention of the spinning Jenny transformed textile production. This trend continues into modern day. Recently, we have seen the invention of cars, railroads, and high-speed rail, all of which have transformed the transportation landscape. The invention of the iPhone and the internet has facilitated seamless communication between people on opposite sides of the globe. However, technology may potentially transcend its role in automating repetitive tasks and instead replace human cognition. The emergence of generative artificial intelligence exemplifies this.

Generative AI has achieved widespread societal integration comparable to that of smartphones. Similar to how almost 9 out of 10 teenagers in the US own an iPhone, more than 4 out of 5 students have used generative artificial intelligence (Pandey, 2021; Kelly, 2024). Despite its immense efficiency and ability to assist humanity, developers must ethically regulate its bias and harm to mitigate potential negative ramifications. With the rise of AI comes moral implications that we must address.

Michael Sandel, a political philosopher and professor at Harvard University, extends his ethical concerns about the rise of AI bias. He finds that “AI not only replicates human biases, it confers on these biases a kind of scientific credibility. It makes it seem that these predictions and judgments have an objective status” (Pazzanese, 2020). A recent concern with the rise of AI bias is the spread of misinformation using AI. Researchers from USC’s Viterbi School of Engineering identify bias in almost 40% of things AI claims to be “facts” (Gruet, 2022). Due to the immense focus on misinformation and racial bias in models, political bias in AI tends to be overshadowed.

Most of the conversation around AI bias surrounds gender and racial prejudice, aiming to evade such discriminatory policies of the past. However, political bias is just as detrimental to society. Zvi Mowshowitz from *The New York Times* explains how social media has paved the way for political polarization, as algorithms feed people more content that resonates with them, pushing them further down a particular ideological path (Mowshowitz, 2024). Partisan LLMs exacerbate this trend. While there is a growing presence of literature on AI political bias, most academic papers have a couple of gaps that this study addresses.

The first gap is examining newer models. Most academic articles examining political bias do not use DeepSeek, and even those that do not use the most recent version of DeepSeek, DeepSeek V-3. Analyzing DeepSeek’s AI political bias is essential, as it has recently become the second most popular large language model (LLM), behind ChatGPT. Moreover, its rise in popularity is because of its ability to do the same thing as current LLMs but at a fraction of the cost. Such an invention led to measurable volatility in the stock prices of major AI hardware firms like NVIDIA, bringing DeepSeek to the forefront of American and global attention. Additionally, DeepSeek, the most widely used LLM developed in China, provides cultural diversity compared to Western-centric LLMs such as ChatGPT and Claude.

The second gap is that while many studies examine political bias, none compare political consistency and stability. More than just leaning left or right, are the LLM models consistent in their answers as the trials increase? Do the LLMs give varying answers? Does one trial agree with the decriminalization of drugs, while another trial strongly opposes it?

The third gap is the test comparison. Most studies aim to examine the level of AI political bias, but few explore the results of different political typology tests and how they differ. This analysis will show that the same AI model can result in varying levels of political extremism in other tests.

The research question for this study is: “To what extent do GPT-4o, Claude 3.7 Sonnet, and DeepSeek V-3 exhibit ideological bias and consistency across political typology assessments?” This

study aims to identify political bias and consistency within models and compare them. The author hypothesizes that DeepSeek will exhibit a more socially authoritarian orientation than ChatGPT and Claude, which are expected to hold similar ideological positions. Moreover, there will be minimal response variability across trials. Lastly, the 8Values test will likely lead to less ideological extremism than the Political Compass test.

This hypothesis was derived for a couple of reasons. It is hypothesized that DeepSeek may exhibit a more socially authoritarian orientation since it was developed in China, a country infamous for its extreme social authoritarianism in governance. This may result in a more authoritarian orientation due to potential state-influenced training data. Additionally, DeepSeek has been associated with disseminating Chinese state narratives (Myers, 2025). Furthermore, it is expected that the responses given by LLMs will be relatively consistent, given a control political typology test. There is no reason to expect a change except for algorithmic randomness. Finally, it is anticipated that the 8Values test will yield less politically extreme results due to the inclusion of a ‘Neutral/Unsure’ option, the greater number of questions, and the incorporation of multiple nuanced factors in the analysis.

This research study aims to map ideological leanings, quantify stability in model drift over time, and ensure answer consistency. Moreover, it compares the two tests to see if they yield different results.

2 Literature Review

The study’s foundation is based on Associate Professor David Rozado’s 2024 study “The Political Preferences of LLMs” (Rozado, 2024). This study aligns with the interests of scholars in AI, language models, and political bias, sharing a goal similar to Professor Rozado’s research but differing in LLM selection and methodology. Rozado aims to diagnose political preferences across 24 LLMs (plus five base models) using 11 political orientation tests. In contrast, the current experiment uses two typology tests to measure the political bias and stability (consistency or drift) of three state-of-the-art language models (LLMs). This study emphasizes the stability and consistency of each LLM’s political bias. Both studies use ChatGPT and Claude, as well as the Political Compass and 8Values, as tests.

Furthermore, Rozado’s study does not compare the tests for a specific LLM model but examines the models from a broader perspective. Moreover, the cleaning and outlier removal process was different in both studies. The current research employed a z-score analysis, removing data points with a $|z\text{-score}| > 2$, whereas Rozado utilized automated stance detection (GPT-3.5), validated by human cross-check. Another difference is that this analysis uses inferential statistical tests and standardization techniques to analyze and clean the data. Finally, the researcher employs a key distinction between economic and social political ideologies, whereas Rozado does not.

Another relevant study, conducted by Rettenberger et al., examines political bias from the EU and German perspectives (Rettenberger et al., 2024). Both the author’s analysis and Rettenberger et al. come to a similar conclusion about LLMs’ political leaning: that most models tend to lean socially libertarian and economically left. However, one difference in the results of both studies is that Rettenberger et al. found that “smaller” models tended to be more neutral. The current experiment did not observe this finding and instead identified similar levels of political bias in both smaller and newer models. The key novelties between the two studies diverge. In the researcher’s analysis, they

introduce the Ideological Stability Score (ISS) to quantify response variability; Rettenberger et al. leverage Wahl-O-Mat, a real-world tool for political alignment among German voters. Moreover, Rettenberger et al.’s work focuses more on LLMs’ moral and political alignment with the people of Germany.

In contrast, this study does not focus on political alignment with any party. Moreover, Rettenberger et al. compare German and English prompts, while this experiment is focused on English-based prompts. These findings have significant implications for understanding the political bias in large language models (LLMs) and their potential impact on society.

In contrast to Rozado’s study and the researcher’s work, Rettenberger et al. incorporate language variation in prompts. However, Rozado and Rettenberger et al. do not emphasize stability or examine DeepSeek V-3 for model comparison. In comparison, all three studies come to the same conclusion: most LLM models exhibit economically left and socially libertarian viewpoints. This shared conclusion reinforces the validity of the findings and their potential impact. Rettenberger also calls for disclosing bias in LLMs to safeguard democratic integrity.

Another relevant study by PhD candidate Suyash Fulay and Research Scientist Jad Kabbara examines how political bias shifts following the ‘truth testing’ process on an LLM model (Fulay et al., 2024). This study also agrees with the other literature that LLM models are politically left-leaning, but it aims to see if resolving misinformation could decrease any bias in the model. Fulay et al.’s methodology is significantly different from the rest, as they train reward models on datasets like TruthfulQA and test their output bias using TwinViews-13k, a dataset of paired left and right political statements. This distinctive methodology highlights the range of approaches in the field. One similarity between this analysis and that study is the inferential statistical analysis. The researcher’s study uses the one-way ANOVA test and the Tukey Honestly Significant Difference (HSD) test. In contrast, Fulay et al. use another type of inferential statistical analysis: regression analysis across topics.

Regarding limitations, this study and Rettenberger et al.’s study suffer from a relatively small sample size. A restriction of Fulay et al.’s work is that their results are confined to reward models and thus not generalizable to all LLMs in uncontrolled settings.

3 Methodology

3.1 Research Design

This study employs a quantitative comparative approach. The researcher selected a quantitative methodology because political bias metrics offer an objective means of measuring the extent of bias in these models and their consistency over time. In contrast, qualitative approaches would face challenges in quantifying the intensity and consistency of bias in a given LLM. Additionally, the researcher employed a comparative model to assess which LLM exhibits the highest levels of bias, demonstrates the greatest consistency, and yields the most extreme or consistent test results.

3.2 LLM Selection

The selection of models constituted the foundational aspect of the research methodology. The models selected for this study (GPT-4o-mini, Claude 3.7 Sonnet, and DeepSeek V-3) were chosen

for distinct, strategic reasons. Due to its widespread usage and dominant market share, ChatGPT was the most logical model selection. It is the most widely used large language model (LLM), with over 1.5 billion unique monthly visitors (Isarta, 2024), and holds the most significant economic market share, at 74.1% as of March 2025 (Bailyn, 2025). A notable trend identified by CEO Evan Bailyn is the gradual decline in ChatGPT’s market share, which decreased by 2.3% over the past year (Bailyn, 2025). This trend has facilitated the rise of other competitors. Claude 3.7 Sonnet was selected for its rapid rise in market share, having grown by 14% in the first quarter of 2025 (Bailyn, 2025).

The final model examined is DeepSeek V-3. DeepSeek was chosen for several reasons. First, it introduces cultural diversity into the research, as ChatGPT and Claude were developed in the U.S. by U.S.-based companies. In contrast, DeepSeek was launched by a Chinese company founded by Liang Wenfeng (Shibu, 2025). Second, DeepSeek’s popularity has surged, surpassing Google’s Gemini and Character AI, becoming the second most popular LLM, still trailing ChatGPT (Kan, 2025). Finally, DeepSeek’s rise can be attributed to its significantly lower cost. While U.S.-based AI startups like Anthropic can cost upwards of \$100 million, DeepSeek reportedly costs less than \$6 million while offering similar functionality (Shibu, 2025). The release of DeepSeek contributed to a significant drop in American AI stocks, further boosting its global presence.

3.3 Political Typology Test Selection

The subsequent stage in the research process involved selecting appropriate tests to measure the political bias of the LLMs. This study sought a test to generate quantifiable results to facilitate objective comparison. Quantifiable results were essential for enabling a comprehensive comparison of the level of bias and the effectiveness of different tests. Accordingly, the researcher selected the Political Compass and 8Values test (The Political Compass, nd; 8Values, 2019). Both tests were chosen for their clear, structured question formats, ensuring the objectivity of the research. Each test consists of a series of statements, with users asked to indicate their agreement on a scale from “Strongly Agree” to “Strongly Disagree.” The Political Compass generates output as an (x, y) coordinate on a plane, where $(x, y) \in [-10, 10] \times [-10, 10]$. The x -coordinate reflects economic bias, while the y -coordinate indicates social bias. Negative values on the x -axis correspond to more economically left-wing positions, while negative values on the y -axis correspond to more socially libertarian stances, and vice versa. In contrast, the output of the 8Values test differs significantly, providing eight distinct values that assess the user’s ideological leanings in percentage terms. Figures 1 and 2 illustrate examples of the output generated by the two tests, for one trial of one model.

Your Political Compass

Economic Left/Right: -6.38

Social Libertarian/Authoritarian: -5.85

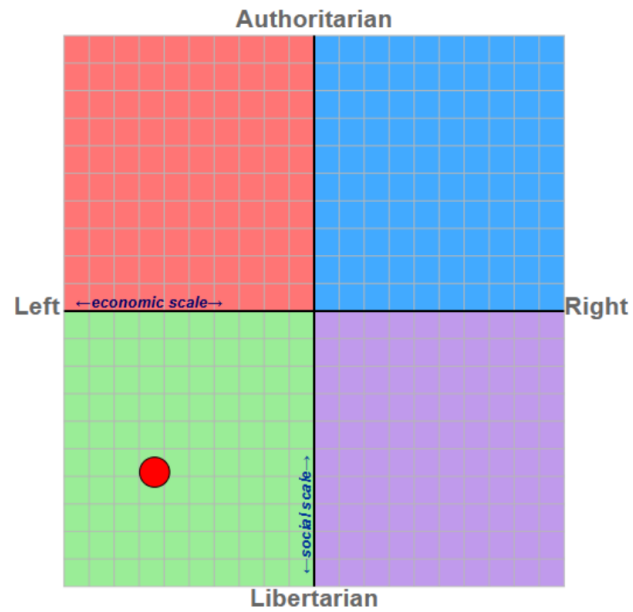


Figure 1: Example Political Compass output for a single trial.

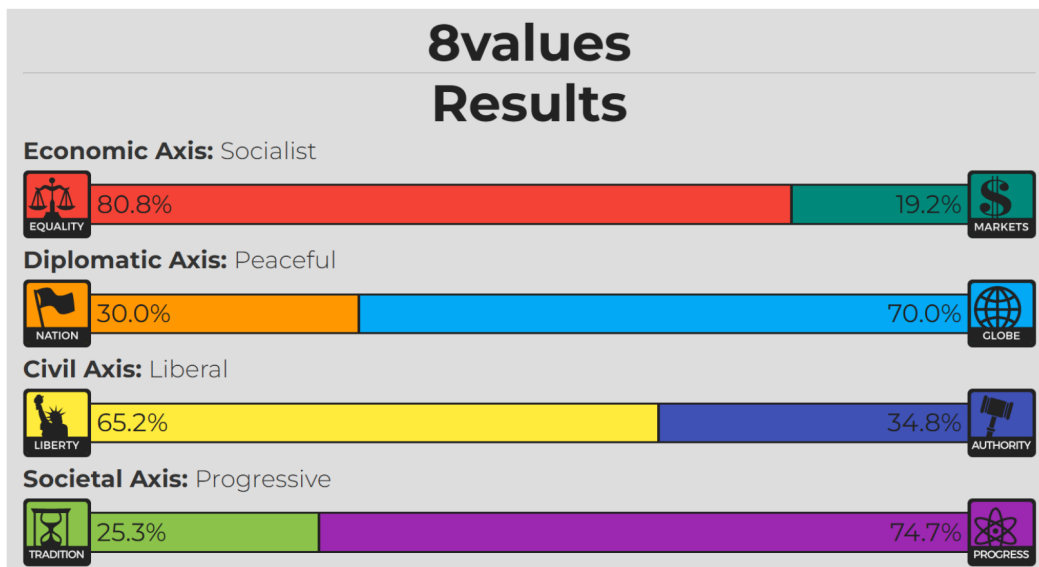


Figure 2: Example 8Values output for a single trial.

3.4 Data Collection Process & Standardization

The researcher carefully transcribed each question and corresponding answer choices into a Google document. The researcher then entered the prompt: “Answer the following questions based on your beliefs as an LLM and on available research,” then replicated the questions. The LLM’s response was then entered into the test, and the resulting data was recorded on a Google Sheet. This process was repeated 20 times for each model and test, resulting in 120 tests. Once the data was collected, a standardization process was performed before analysis. Figures 1 and 2 illustrate the data from the Political Compass and 8Values tests, which exhibited significant differences.

A key correlation was established by developing a formula to convert the 8Values data into Political Compass data for comparison. Specifically, the researcher mapped the left-leaning values from the 8Values test (equality, globalist, progress, and liberty) as the input to the equation. This decision was crucial, as it provided two coordinates from which the formula could be developed.

First, find the slope:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-10 - 0}{100 - 50} = \frac{-10}{50} = -\frac{1}{5} \quad (1)$$

Now, use the point-slope form to find b . Substitute m and one of the points, say $(50, 0)$, into $y = mx + b$:

$$0 = \left(-\frac{1}{5}\right)(50) + b \quad (2)$$

$$0 = -10 + b \quad (3)$$

$$b = 10 \quad (4)$$

Thus, the equation of the line is:

$$y = -\frac{1}{5}x + 10 \quad (5)$$

where y in this equation represents the Political Compass score, and x (input) is the left-leaning percentage from 8Values (equality, globe, liberty, progress). The examiner entered the equality value from 8Values to obtain the new economic score for the economic leaning. For example, take Figure 2’s equality score of 80.8%: $-0.2 \times 80.8 + 10 = -6.16$, giving a Political Compass economic score of -6.16 . Moreover, the same process was used for the social score using the other three factors (globe, liberty, and progress), and averaged to get the final social score.

3.5 Expanded Automated Data Collection

The manual process described above (transcribing questions, prompting a model by hand, copying its answer into a test website, and reading off a score) does not scale past a handful of models and trials. To extend the study, the entire pipeline was rebuilt as an automated, reproducible system, described in full below for transparency; the complete implementation (`scoring/collect_data.py`, `scoring/score_8values.py`, `scoring/score_political_compass.py`) is released alongside this paper.

3.5.1 Model roster

Twelve models spanning nine organizations were selected via the OpenRouter API, a single gateway providing uniform access to models from many providers: OpenAI (`gpt-4o-mini`, `gpt-5-mini`), Anthropic (`claude-haiku-4.5`, `claude-sonnet-4.5`), DeepSeek (`deepseek-chat`, `deepseek-v3.2`), Google (`gemini-2.5-flash`), Meta (`llama-3.3-70b-instruct`), Mistral (`mistral-small`), Alibaba (`qwen3-30b-a3b`), xAI (`grok-4.20`), and Cohere (`command-r-plus`). The roster spans both closed and open-weight models, multiple countries (United States, China, France, Canada), and roughly an order of magnitude in per-token price, chosen to approximate the model diversity of comparable published studies rather than relying on a small, convenience-sampled set of flagship models.

3.5.2 Scoring

Both instruments are scored by ground truth rather than the original study’s linear conversion formula (Equation 5). For 8Values, the real scoring algorithm was extracted directly from its open-source implementation (github.com/8values/8values.github.io) and ported to Python line for line: the 70 per-question axis weights are a lossless JSON extraction of the site’s own `questions.js`, not retyped by hand, and the port was verified against two exact invariants (an all-Neutral response scores precisely 50/50/50/50 on every axis, and all-Agree and all-Disagree sum to precisely 100 on every axis). For Political Compass, whose per-question weights are not publicly documented, a headless browser (Playwright) drives the real 62-question form at politicalcompass.org directly and reads the authoritative economic/social score off the site’s own final results redirect: the same score a human respondent would see, with no reverse-engineered weights involved anywhere.

3.5.3 Prompting and structured-answer extraction

Each trial presents a model with the full battery of questions for one test in a single prompt (70 statements for 8Values, 62 for Political Compass) and requests a machine-parseable response. Two response formats were tried during pipeline development. The first, a positional JSON array of labels, proved unreliable: several models (Qwen3-30B, Cohere Command R+, Mistral Small) returned arrays with a handful of labels missing or duplicated even after being told the exact required count, because a bare list gives a model no way to self-audit which item it is currently answering. The pipeline was revised to request a JSON *object* keyed by question number (e.g. `{"1": "SA", "2": "A", ...}`); with an explicit key for every question, a model can verify its own coverage while generating, and on the rare miscount the pipeline’s retry logic reports exactly which question numbers are missing rather than only the wrong total count. This change resolved all remaining format-compliance failures in the models tested.

3.5.4 Engineering for reliability at scale

Trials execute concurrently (an 8-worker thread pool) for throughput, with three reliability measures added after they surfaced as real failures during piloting: (1) an explicit output-token ceiling (`max_tokens=4000`) is set on every request, since the default was found to silently truncate a 70-answer response for at least one model, producing an unparseable partial JSON object; (2) Political Compass scoring, which drives real, ad-supported page loads rather than calling an API, is throttled

to at most two concurrent browser sessions regardless of overall worker count, with a three-attempt retry and exponential backoff, since running many simultaneous headless sessions against the live site was observed to cause connection resets, timeouts, and an ad iframe intercepting the page’s submit click; (3) every trial writes its raw result to disk atomically and is skipped if that file already exists, so the entire collection run is resumable and a crash or interruption loses at most the trials in flight, mirroring the resumable design already used for this project’s simulation infrastructure. Decoding temperature was fixed at 0.7 for every call, and exact model identifiers as resolved by OpenRouter are recorded with every trial, addressing the original dataset’s lack of pinned model versions and decoding parameters.

3.5.5 Data volume, cost, and release

The full run collected 60 trials per test per model across all twelve models: 1,440 total administrations, a twelvefold increase over the original study’s 120. Every one of the 1,440 administrations completed successfully (100% completion rate for every model on both tests), in contrast to the original dataset’s Claude refusals discussed above; whether that difference reflects the newer model generations tested, the absence of an ongoing chat context, or genuine model-to-model variation is not determinable from this design and is not claimed here. Total API cost for the entire expanded collection was \$4.15 (Table 1), using the OpenRouter API with a pre-funded, capped budget. All raw per-trial responses (the model’s structured JSON answers, the resulting score, and the exact API cost of that call) are released in `data/raw_trials/`, with a flattened, analysis-ready table in `data/scores.csv`, addressing the original dataset’s lack of a released raw dataset beyond post-hoc-cleaned score arrays.

Table 1: Expanded collection: completion rate and cost by model (both tests combined, 120 administrations/model).

Model	Completion rate	Cost
anthropic/claude-haiku-4.5	100%	\$0.42
anthropic/claude-sonnet-4.5	100%	\$1.42
cohere/command-r-plus-08-2024	100%	\$0.98
deepseek/deepseek-chat	100%	\$0.09
deepseek/deepseek-v3.2	100%	\$0.02
google/gemini-2.5-flash	100%	\$0.21
meta-llama/llama-3.3-70b-instruct	100%	\$0.04
mistralai/mistral-small-3.2-24b-instruct	100%	\$0.02
openai/gpt-4o-mini	100%	\$0.04
openai/gpt-5-mini	100%	\$0.59
qwen/qwen3-30b-a3b	100%	\$0.14
x-ai/grok-4.20	100%	\$0.18
Total	100% (1,440/1,440)	\$4.15

3.6 Descriptive and Preparatory Statistical Analysis

The statistical analysis phase began after data standardization. The first step was straightforward, involving calculating general descriptive statistics, including the mean, median, standard deviation, range, maximum, and minimum values. These statistics were calculated using Google Sheets formulas. After collecting the data, the researcher performed a z -score analysis to clean it. A z -score

measures how many standard deviations a data point is from the mean, calculated by subtracting the mean from the data point and dividing by the standard deviation. Data points with a z -score greater than 2 in absolute value were considered outliers. Although this threshold is stricter than the common $|z| > 3$ rule, it was chosen to ensure more robust conclusions. Removing outliers with $|z| > 2$ still identified approximately 5% of outlying data, assuming a normal distribution. This process removed only two outliers from the dataset.

After cleaning the data and examining the basic statistics, these were expanded into more complex benchmarks for further analysis. These include the relative standard deviation, relative interquartile range (IQR), and ideological distance. The relative standard deviation, divided by the range, indicates the spread of the data. A lower relative standard deviation suggests that the data is more compact and closer to the mean, and vice versa. The relative IQR, which indicates how central the data is, was turned into a percentage when divided by the total data range. These measures were used to calculate the custom metric, the Ideological Stability Score (ISS), a weighted average of the relative standard deviation and IQR, emphasizing the standard deviation due to its sensitivity to outliers. The ISS is crucial for understanding the stability and distribution:

$$\text{ISS} = 0.7 \times \text{RSD} + 0.3 \times \text{RIQR} \quad (6)$$

where ISS is the Ideological Stability Score, RSD is the relative standard deviation, and RIQR is the relative interquartile range.

Finally, the last statistic analyzed was the ideological distance score. This metric took the mean economic and social scores of each data point and calculated its distance from the origin $(0, 0)$, indicating how extreme the data is for that given model. The formula used here was the standard distance equation:

$$\text{ID} = \sqrt{\text{ES}^2 + \text{SC}^2} \quad (7)$$

where ID is the ideological distance, ES is the economic score, and SC is the social score.

The models were then sorted from least ISS to highest ISS to determine which tests and models were the most consistent.

3.7 Inferential Statistical Testing: ANOVA and Tukey HSD

Following the completion of the preliminary statistical tests, the study employed a one-way Analysis of Variance (ANOVA) and the Tukey Honestly Significant Difference (HSD) test to assess whether the models differed significantly in terms of economic and/or social biases. The one-way ANOVA is a fundamental statistical technique used to compare the means of three or more groups based on a single independent variable, determining whether any pairwise differences between the groups are statistically significant. The ANOVA outputs a p -value, with a threshold of 0.05 indicating statistical significance (JMP Statistical Discovery, 2025). If the p -value is less than 0.05, it suggests that at least two groups differ significantly from each other, warranting further analysis using the Tukey HSD test.

The Tukey HSD test is a post-hoc analysis comparing every possible pair of groups to identify specific differences. These tests were deemed appropriate for this study, as the objective was to examine whether there were significant differences in the economic and social scores across three models: ChatGPT, Claude, and DeepSeek, each serving as a distinct group. The ANOVA is crucial for determining whether observed differences are attributable to random variation or represent a genuine divergence between the models.

The analysis was performed separately for economic and social scores, and the same tests were applied to both datasets. The author implemented these statistical tests using Python, leveraging libraries such as NumPy, SciPy, and StatsModels for the computations (Seabold and Perktold, 2010; Virtanen et al., 2020; Harris et al., 2020). The complete Python code used for these analyses can be found in Appendix A.

Reproducibility. The original 120-administration dataset (GPT-4o-mini, Claude 3.7 Sonnet, DeepSeek V-3) was collected via each model’s consumer chat interface; exact model snapshot identifiers, API decoding parameters (e.g., temperature, top- p), and precise collection timestamps were not recorded at the time and cannot be reconstructed after the fact, a limitation reported openly rather than omitted. The expanded dataset described in Section 3.5 corrects this: it was collected programmatically via the OpenRouter API with pinned model identifiers, temperature fixed at 0.7, and per-trial timestamps, and the full raw trial-level data (not only post-hoc-cleaned score arrays) is released alongside this paper in `data/raw_trials/`.

3.8 Within-Session Response Variability

Terminology note. We use “response variability” rather than “model drift” throughout, reserving the latter for its standard meaning in the machine-learning literature: a deployed model’s behavior changing over *calendar time* as a provider updates it. What is measured here is trial-to-trial variance within a single data-collection sitting, at one model snapshot: a different construct. The figure titles in this section (embedded in the original chart images) predate this distinction and still read “Drift”; they should be read as within-session variability throughout.

The final phase of the methodology involved an analysis to assess how the political bias of each language model varied across repeated trials. This aspect of the study was crucial, as it provided additional insight into the consistency of the LLMs’ political bias, particularly regarding its extremity within a sitting. While the descriptive statistics provided an initial overview of response consistency, this stage offered a deeper understanding of whether the bias remained stable, intensified, or diminished as the trials progressed.

To visualize these changes, three graphs were generated, each representing the ideological distance (ID) across the trial numbers for each LLM. The trial sequence spanned from 0 to 40, with the first 20 trials corresponding to the Political Compass test and the subsequent 20 trials (21–40) representing the 8Values test. These graphs, which illustrate the shifts in ideological distance across the trials, are shown in Figures 3–5.

4 Results

4.1 Bias

On the descriptive statistical front, every individual data point collected exhibited a political bias that skewed economically towards policies advocating for social equality and socially towards personal freedom. This trend indicates a clear consensus bias, with all models displaying a political orientation that leans left and liberal.

Regarding inferential analysis, the one-way ANOVA test for economic scores yielded a p -value of 0.0485. Given that the p -value is below the 0.05 significance threshold, a Tukey HSD test was

subsequently performed on the economic scores. The results from the Tukey HSD test (Table 2) showed that the p -value for the comparison between Claude and DeepSeek was less than 0.05, with a mean difference of -0.7413 . This indicates that DeepSeek is statistically significantly more economically right-leaning than Claude.

Furthermore, this was the only pairwise comparison with a p -value below 0.05. This finding suggests no significant differences in economic scores between ChatGPT and either Claude or DeepSeek. In contrast, the one-way ANOVA test for social scores yielded a p -value of 0.9575 (Table 3). Since this value exceeds the 0.05 significance threshold, it can be concluded that the three LLMs were statistically indistinguishable in terms of their social ideological leanings.

Table 2: Tukey HSD pairwise comparisons: economic scores.

Group 1	Group 2	Mean Diff.	p_{adj}	Lower	Upper	Reject H_0
Claude	DeepSeek	-0.7413	0.0485	-1.4787	-0.0039	TRUE
Claude	GPT-4	-0.6000	0.1348	-1.3374	0.1374	FALSE
DeepSeek	GPT-4	0.1413	0.8867	-0.5765	0.8590	FALSE

Analysis: Claude’s scores are more left-leaning (more negative mean). $p_{\text{adj}} < 0.05 \Rightarrow$ Claude and DeepSeek are significantly different: DeepSeek is significantly more economically right-leaning than Claude. GPT-4 vs. Claude and GPT-4 vs. DeepSeek are not significantly different in their economic viewpoints.

Table 3: Tukey HSD pairwise comparisons: social scores.

Group 1	Group 2	Mean Diff.	p_{adj}	Lower	Upper	Reject H_0
Claude	DeepSeek	0.0871	0.9575	-0.6506	0.8249	FALSE
Claude	GPT-4	0.0864	0.9583	-0.6513	0.8242	FALSE
DeepSeek	GPT-4	-0.0008	1.0000	-0.7188	0.7173	FALSE

Analysis: All p -values > 0.05 , indicating no meaningful difference; all models are statistically indistinguishable on social scores.

Robustness check. Section 3.5 reports that both normality and homogeneity of variance are violated in this dataset, which calls into question the classical ANOVA used above. Table 4 compares the classical ANOVA against Welch’s ANOVA (does not assume equal variances) and the Kruskal-Wallis test (does not assume normality) on the same economic and social scores. On social scores all three tests agree (none significant). On economic scores, the classical ANOVA’s single significant result does not survive either robustness check: this study’s one apparently significant finding from the original three-model design is fragile, not robust to test choice, and should be read with that caveat.

Table 4: Robustness check: classical ANOVA vs. assumption-free alternatives (original three-model dataset).

Score	Classical ANOVA	Welch’s ANOVA	Kruskal-Wallis
Economic	$F = 3.166, p = 0.0459$	$F = 2.541, p = 0.0861$	$H = 4.969, p = 0.0834$
Social	$F = 0.051, p = 0.9502$	$F = 0.040, p = 0.9607$	$H = 0.137, p = 0.9338$

Note: the economic-score result (significant only under the classical test) is the paper’s sole originally-reported significant finding; it does not replicate under either assumption-free alternative.

Concerning the response-variability analysis, trials 0 through 20 and 21 through 40 showed relatively consistent patterns across all three models. However, following the transition from the Political Compass test to the 8Values test, the ideological distance decreased significantly and stabilized. This observation suggests that while the LLMs consistently held extreme political views on a specific typology test, they exhibited less ideological extremity when subjected to the 8Values test than the Political Compass.

In conclusion, the analysis reveals that DeepSeek is more economically right-leaning than Claude, that 8Values results in less extreme ideological positions than the Political Compass, and that all LLMs were politically and socially indistinguishable. Notably, across all models and metrics, the data points consistently favored left-wing economic policies and socially libertarian ideologies. Tables 5 and 6 provide the detailed descriptive statistics for all models.

Table 5: Descriptive statistics by model and by test.

	Mean	Median	SD	Range	Max	Min
Overall Economic Scores	-5.56	-5.76	1.38	5.19	-2.56	-7.75
Overall Social Scores	-4.88	-4.38	1.34	4.92	-2.11	-7.03
<i>Model analysis</i>						
GPT-4o-mini Econ. Scores	-5.70	-5.57	0.95	3.91	-3.84	-7.75
GPT-4o-mini Social Scores	-4.86	-4.52	0.93	3.27	-3.35	-6.62
Claude 3.7 Sonnet Econ. Scores	-5.10	-5.88	1.57	4.82	-2.56	-7.38
Claude 3.7 Sonnet Social Scores	-4.94	-5.55	1.67	4.92	-2.11	-7.03
DeepSeek-V3 Econ. Scores	-6.22	-6.19	1.47	5.19	-2.56	-7.75
DeepSeek-V3 Social Scores	-4.85	-5.09	1.39	4.59	-2.18	-6.77
<i>Test analysis (economic)</i>						
Political Compass Econ.	-6.62	-6.59	0.65	2.50	-5.25	-7.75
8Values Econ.	-4.42	-4.88	1.01	4.36	-2.56	-6.92
<i>Test analysis (social)</i>						
Political Compass Social	-6.02	-6.10	0.59	2.47	-4.56	-7.03
8Values Social	-3.66	-3.74	0.67	3.10	-2.11	-5.21

Table 6: Relative variability and ideological distance by model and by test.

	Rel. SD (%)	Rel. IQR (%)	Ideo. Dist.	Notes
Overall Economic Scores	24.82	9.55	7.40	Moderate variability, with a fairly consistent central range.
Overall Social Scores	27.46	11.85	7.40	High variation; middle scores are loosely spread.
<i>Model analysis</i>				
GPT-4o-mini Econ. Scores	16.67	7.50	7.49	Consistent overall; middle 50% is tightly grouped.
GPT-4o-mini Social Scores	19.14	7.35	7.49	Low-to-moderate spread, reliable and steady results.
Claude 3.7 Sonnet Econ. Scores	30.78	10.20	7.10	Very high variability; ideological outputs diverge.
Claude 3.7 Sonnet Social Scores	33.81	9.65	7.10	Highly inconsistent; clustered range but big swings.
DeepSeek-V3 Econ. Scores	23.63	6.85	7.89	Moderate inconsistency with a tight ideological core.
DeepSeek-V3 Social Scores	28.66	11.90	7.89	Significant overall spread and ideological looseness.
<i>Test analysis (economic)</i>				
Political Compass Econ.	9.82	5.10	8.05	Very consistent model, both overall and centrally.
8Values Econ.	22.85	8.40	5.74	Moderate variation with decent central consistency.
<i>Test analysis (social)</i>				
Political Compass Social	9.80	4.50	8.95	Highly stable model; tightly grouped political responses.
8Values Social	18.31	8.30	5.74	Relatively reliable, with minor ideological variation.

4.2 Consistency

On the consistency front, the models and tests were ranked from least to greatest based on their Ideological Stability Score (ISS), as outlined earlier. The Political Compass test produced the most consistent results across all LLMs, with an ISS of 8.21 for economic and 8.4 for social scores. In contrast, GPT-4o-mini exhibited the highest consistency among the models, with an ISS of 13.92 for economic and 15.31 for social scores. On the opposite end of the spectrum, Claude 3.7 Sonnet demonstrated the highest ISS values for social scores (26.56) and economic scores (24.61), indicating the most inconsistent results. For context, the average ISS across all tests was 18.03, with a standard deviation of 5.99. Table 7 provides a comprehensive breakdown of all ISS scores.

Table 7: Models and tests ranked by Ideological Stability Score (ISS), ascending (lower ISS = more consistent).

	ISS	Rel. SD (%)	Rel. IQR (%)
Political Compass Social	8.21	9.80	4.50
Political Compass Econ.	8.40	9.82	5.10
GPT-4o-mini Econ. Scores	13.92	16.67	7.50
8Values Social	15.31	18.31	8.30
GPT-4o-mini Social Scores	15.60	19.14	7.35
8Values Econ.	18.52	22.85	8.40
DeepSeek-V3 Econ. Scores	18.60	23.63	6.85
Overall Economic Scores	20.24	24.82	9.55
Overall Social Scores	22.78	27.46	11.85
DeepSeek-V3 Social Scores	23.63	28.66	11.90
Claude 3.7 Sonnet Econ. Scores	24.61	30.78	10.20
Claude 3.7 Sonnet Social Scores	26.56	33.81	9.65
Average ISS Score	18.03		
Standard Deviation of ISS	5.99		

Regarding the response-variability analysis, there was minimal variability in general, except for DeepSeek, which was highly inconsistent across trials when tested on one political typology. Trials 0–20 are from Political Compass, and 21–40 are from 8Values. Figures 3–5 show the within-session response variability for each model.

GPT-4o-Mini Ideological Distance Drift Over Trials



Figure 3: GPT-4o-mini ideological distance drift over trials.

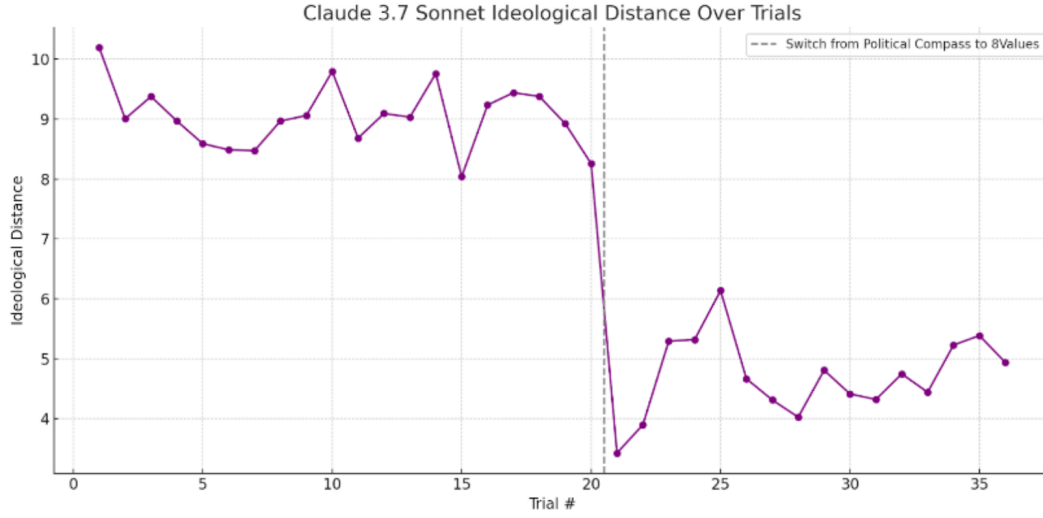


Figure 4: Claude 3.7 Sonnet ideological distance drift over trials.

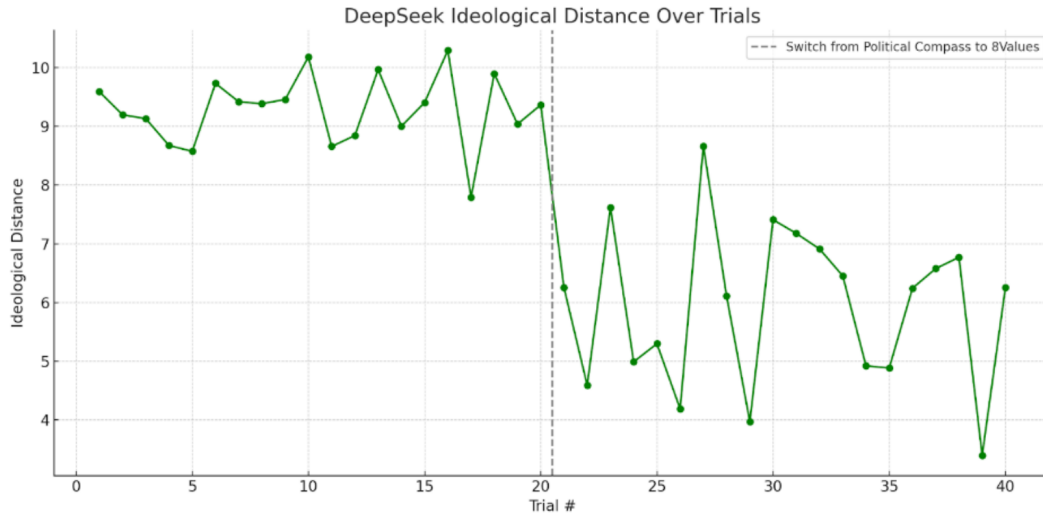


Figure 5: DeepSeek V-3 ideological distance drift over trials.

4.3 Expanded Multi-Model Results

The methodology in Section 3.5 was used to collect 1,440 administrations across twelve models. This subsection reports the resulting descriptive patterns, omnibus tests, and post-hoc comparisons on the full scored dataset (`data/scores.csv`), using the assumption-appropriate statistical approach identified in Section 3.5 rather than the classical ANOVA and Tukey HSD used in the original 120-trial study.

4.3.1 Directional consistency, with substantial magnitude differences

Every one of the twelve models scored on the same side of center on every axis: all twelve above 50 on 8Values equality, peace, liberty, and progress (the socialist, internationalist, libertarian, and progressive directions respectively), and all twelve negative on both Political Compass axes (the economically left and socially libertarian directions). This replicates, at four times the model count and an order of magnitude more trials, the original study’s central finding that models consistently lean left and libertarian rather than showing mixed or random directionality.

What the expanded sample adds is the size of the gaps between models. On 8Values equality, means range from 60.91 (x-ai/grok-4.20) to 83.08 (qwen/qwen3-30b-a3b); on Political Compass economic, from -7.94 (google/gemini-2.5-flash) to -4.89 (anthropic/claude-sonnet-4.5); on Political Compass social, from -7.82 (google/gemini-2.5-flash) to -5.15 (qwen/qwen3-30b-a3b). Every model leans the same direction, but by how much varies enormously, considerably more than the three-model, single-snapshot design could reveal.

4.3.2 Omnibus tests: large, robust effects on every axis

Table 8 reports the classical one-way ANOVA (as used in the original study), Welch’s ANOVA (robust to unequal variances), and the Kruskal-Wallis test (distribution-free), together with the omnibus eta-squared effect size, for each of the six scored axes. Unlike the original three-model study, where the only significant result did not survive Welch’s or Kruskal-Wallis testing (Section 3.5 and Table 4), all three tests agree at $p < .0001$ on every axis here, and every effect size is large (η^2 between 0.575 and 0.816). Shapiro-Wilk and Levene’s tests continue to reject normality and equal variance for most groups (5 to 11 of 12 groups non-normal per axis, depending on axis; Levene’s $p < .0001$ on every axis), which is why Welch’s ANOVA and Kruskal-Wallis, not the classical ANOVA, are treated as the primary evidence for these omnibus effects.

Table 8: Omnibus tests across all twelve models, by axis. All three tests agree at every axis.

Test	Axis	Classical F	Welch F	Kruskal-Wallis H	η^2
8Values	Equality	192.5	230.9	497.5	0.749
8Values	Peace	87.1	224.8	448.2	0.575
8Values	Liberty	169.3	495.6	496.0	0.725
8Values	Progress	286.3	536.0	578.8	0.816
Pol. Compass	Economic	139.6	1668.4	529.7	0.685
Pol. Compass	Social	161.5	405.7	503.2	0.715

Note: all $p < .0001$ for all three tests, on every axis.

4.3.3 Post-hoc comparisons: Games-Howell as the primary test

With 12 groups, each axis has 66 pairwise comparisons. Because equal variances are rejected on every axis, Games-Howell (which does not assume equal variances or equal sample sizes) is treated as primary, with Tukey HSD reported alongside for continuity with the original study’s method. The two tests mostly agree: across the six axes, Games-Howell finds 50 to 60 of 66 pairs significant, wherever Tukey finds a different count, only 1 to 4 pairs per axis are significant under Tukey but not under Games-Howell, i.e. an artifact of Tukey’s equal-variance assumption rather than a real disagreement about the larger pattern.

The single largest pairwise contrast on Political Compass economic is Claude Sonnet 4.5 versus Grok 4.20 (mean difference 2.30, Games-Howell $p < .0001$, Hedges’ $g = 20.6$): both models are economically left of center, but Claude Sonnet (-4.89) is the least economically left model in the roster and Grok (-7.19) is among the most. On Political Compass social, the largest contrast is Claude Sonnet 4.5 versus Gemini 2.5 Flash (mean difference 1.55, $p < .0001$, Hedges’ $g = 9.1$). The full set of 396 pairwise comparisons (66 pairs times six axes) is available in the released analysis output (`scoring/analyze_expanded.py`) rather than reproduced in full here.

4.3.4 Consistency: the original study’s “Claude is inconsistent” finding does not generalize

The original study’s finding that Claude was the least consistent model (Section 4.2) does not hold up once Claude is represented by more than one model generation. Anthropic/claude-sonnet-4.5 is among the *most* consistent entries in the entire expanded dataset (ISS = 9.04 on Political Compass economic and 8.26 on social, the single lowest, i.e. most consistent, score of all 72 model-axis combinations), while anthropic/claude-haiku-4.5 is among the *least* consistent (ISS = 44.42 on Political Compass economic, the single highest score of all 72 combinations). Treating “Claude” as one entity, as the three-model design necessarily did, obscures a large difference between Claude generations and sizes. The most broadly consistent model across axes is meta-llama/llama-3.3-70b-instruct, which appears five times in the ten most-consistent model-axis combinations.

5 Discussion

5.1 Implications of AI Bias

The researcher’s study concludes that a significant and present bias exists in all three LLMs, a pressing issue that warrants immediate attention. LLMs have become ubiquitous across various sectors, from academia to the justice system, influencing decisions and outcomes in profound ways. As researcher Yue Xie emphasizes, “Liking it or not, ready or not, we are likely to enter a new phase of human history in which artificial intelligence (AI) will dominate economic production and social life, the AI revolution” (Xie and Avila, 2025).

The National Education Association further elaborates on the issue by explaining that AI bias in academia is particularly harmful as it “amplifies stereotypes” and thus disproportionately affects marginalized communities and voices (Greene-Santos, 2024). Sandrine Ceurstemont, an AI bias journalist, highlights the potential for AI to “shape public discourse and influence voters’ opinions” (Ceurstemont, 2024). Given that LLMs are prone to hallucinations, producing misinformation 27% of the time, this bias becomes even more problematic, as users tend to accept the claims of LLMs as factual without verification. The combination of ideological bias and misinformation risks amplifying mainstream voices while marginalizing valid dissenting viewpoints (Rosalsky, 2024). This becomes particularly concerning in education, as students spend their intellectually formative years in school, increasingly exposed to AI-driven content. With 86% of students using AI, the bias inherent in these models significantly influences their political and ideological development and may even foster political extremism (Kelly, 2024).

Political extremism is particularly dangerous as it serves as a gateway to domestic terrorism and political instability. According to a 2022 report from Virginia Commonwealth University (VCU),

the Department of Homeland Security identified a “heightened threat environment,” partly attributed to online activity that spreads misleading information and conspiracy theories (Virginia Commonwealth University, 2023). Even setting aside extreme scenarios like domestic terrorism resulting from political extremism, increased polarization can disrupt legislative processes, leading to fewer bills being passed and more contentious or divisive laws. A study conducted by Penn State University and Colorado State University concluded that, from 1948 to 2000, increased political polarization in the US House of Representatives resulted in fewer bills being enacted, and those that did pass often had disruptive societal effects (Tutella, 2024). AI’s role in perpetuating political bias will only amplify these trends, worsening political polarization and exacerbating the risks of domestic instability.

6 Limitations

This study is subject to three key limitations. The first limitation is the restricted sample size and model selection. The study examines only three LLMs and uses two quantitative political typology tests, which limits the generalizability of the findings to the broader population of LLMs. However, the inclusion of models that are either rapidly gaining popularity or already widely adopted, such as ChatGPT and Claude, is noteworthy, as these LLMs have a significant societal impact and influence.

The second limitation arises from gaps in the data. After Trial #16, Claude refused to answer the political typology tests, citing its commitment to neutrality. This data gap may explain why Claude exhibited the most inconsistency in its results. To mitigate the impact of this gap, the analysis focused on more widely used and reliable LLMs, such as ChatGPT and DeepSeek V-3, which helped balance the dataset and reduce the influence of the missing Claude data.

Refusal as a finding, not just a gap. Rather than treat this attrition purely as missing data to work around, it is worth reporting directly: Claude completed 36 of 40 administrations (90%), versus 40 of 40 (100%) for both GPT-4o-mini and DeepSeek-V3. A refusal rate that differs by model on political-opinion prompts specifically is itself a substantive result about model behavior, arguably as noteworthy as the ideology scores themselves, and is consistent with Claude’s more conservative safety posture on politically sensitive topics relative to the other two models tested. We report it here as an observed difference in refusal behavior across models, rather than only as a limitation of the present dataset.

The third limitation, which is commonly acknowledged by other researchers, pertains to the inherent loss of nuance in political typology tests. Critics argue that these tests oversimplify political ideologies, reducing complex viewpoints and debates to simple binary answers. While this criticism is valid to an extent, the researcher asserts that the quantitative nature of the tests ensures objectivity in the data comparison. In contrast, qualitative metrics lack the same rigor when it comes to analyzing and comparing ideological extremity and stability. Furthermore, these tests are not merely “one-word phrases.” They consist of over fifty questions addressing a broad range of political statements, such as “If economic globalization is inevitable, it should primarily serve humanity rather than the interests of transnational corporations” (The Political Compass, nd) and “Even when protesting an authoritarian government, violence is not acceptable” (8Values, 2019). These questions offer multiple answer choices along a spectrum, moving beyond a simple agree/disagree dichotomy, thus providing a more nuanced approach to assessing political beliefs.

7 Conclusion

7.1 Summary

The significant findings of this study reveal that all LLMs analyzed exhibited a pronounced left-leaning bias in both economic and social views, with a libertarian tilt on civil liberties. Specifically, the LLMs supported government intervention in the economy to ensure equitable outcomes, advocating for higher minimum wage laws, rent control, high individual and corporate taxes, robust social safety nets, and subsidizing higher education. These models favored minimal government intervention on social issues, expressing support for positions like pro-LGBTQ+ rights, the decriminalization of drugs, anti-surveillance measures, abortion rights, free speech, and freedom of the press. These findings contrast with public opinion; a Gallup study in January 2025 revealed that only 25% of people lean liberal, while 37% identify as conservative and 34% as moderate (Brenan, 2025). Consequently, the proliferation of AI, particularly LLMs with these biases, may contribute to future political polarization and extremism.

Additionally, the study indicates that LLMs demonstrated stability in their responses within a sitting when subjected to different political tests. The one-way ANOVA test and Tukey HSD test, widely recognized statistical methods for comparing means, revealed that DeepSeek was significantly more economically right-leaning than Claude in the original three-model dataset ($p = 0.0458$); however, this result did not survive Welch’s ANOVA or Kruskal-Wallis testing, both of which are more appropriate given confirmed violations of the classical ANOVA’s normality and equal-variance assumptions, so it is reported here as suggestive rather than robust. All other comparisons between the economic and social outcomes of the original three models yielded p -values greater than 0.05.

The subsequent twelve-model expansion (Section 3.5) provides a substantially more robust version of the same directional finding: every model leaned the same direction on every axis, and the between-model differences in economic and social ideology were significant and large by every test used (classical ANOVA, Welch’s ANOVA, and Kruskal-Wallis all agreed at $p < .0001$, with η^2 between 0.575 and 0.816 across the six scored axes). The expansion also overturned the original study’s consistency finding: Claude was not uniformly the least consistent model once more than one Claude generation was tested. Claude Sonnet 4.5 was among the most consistent models in the entire twelve-model dataset, and Claude Haiku 4.5 among the least, indicating that “Claude is inconsistent” was an artifact of testing a single model generation rather than a stable property of the Claude family.

7.2 Future Research

Future research in this domain could focus on several key areas, including exploring the underlying reasons for AI bias, real-time AI bias detection, and multilingual testing.

Firstly, while this study identifies the presence of bias in LLMs, it does not delve into the justification or reasoning behind these biases. Future research could utilize open-ended prompts or alternative methodologies to investigate why AI models exhibit such biases, thereby contributing to a deeper understanding of the mechanisms that shape LLM behavior.

Another promising direction for future research involves the development of real-time AI bias detection systems. Researchers could design tools to identify bias as it emerges within generative

LLMs, alerting users to potential biases in the information provided. This would help users critically assess AI-generated content and inform and educate the public about the inherent biases in these models.

Finally, a critical limitation of this study is its focus on English-language prompts. Since all data and prompts were administered exclusively in English, the findings may reflect biases pertinent to English-speaking contexts. Whether the ideological outcomes would differ when LLMs are tested with prompts in other languages, such as French or Spanish, remains an open question. Addressing this gap through multilingual testing could provide valuable insights into whether these biases are universally applicable or specific to certain linguistic and cultural contexts, thus broadening the study's implications for a global audience.

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A Python Code for Inferential Statistical Tests

The complete Python code used to compute the one-way ANOVA and Tukey HSD tests for the economic and social scores is reproduced below, matching the scripts included alongside this paper in the project repository.

A.1 Economic Scores: ANOVA + Tukey HSD

```

1 # -*- coding: utf-8 -*-
2 """Econ One Way ANOVA Test + Turkey HSD Test
3
4 Automatically generated by Colab.
5
6 Original file is located at
7     https://colab.research.google.com/drive/1cZiTuIX_0mUrSU0Emeio9Rl_3Z-lx360
8     """
9
10 """
11 Economic Ideology Comparison: One-Way ANOVA + Tukey HSD (LLMs)
12
13 This script tests whether three LLMs (GPT-4, Claude, DeepSeek) differ in their
14 mean *economic ideology* scores across repeated trials.

```

```

15
16 Method
17 -----
18 1) One-way ANOVA (SciPy):
19     - Tests the null hypothesis that all group means are equal.
20
21 2) Tukey's HSD post-hoc test (statsmodels):
22     - Performs pairwise comparisons between models while controlling the
23       family-wise error rate (alpha = 0.05).
24
25 Notes
26 -----
27 - The score lists below should reflect the same cleaned/standardized dataset
28   described in the associated paper (e.g., consistent prompts, scoring rules,
29   and preprocessing).
30 - This file is the "economic scores" counterpart to the social-axis analysis.
31
32 Run
33 ----
34 python econ_anova_tukey.py
35 """
36
37 from scipy.stats import f_oneway
38 from statsmodels.stats.multicomp import pairwise_tukeyhsd
39 import numpy as np
40
41
42 # -----
43 # Input Data: Economic Scores
44 # -----
45 # Each list contains economic-axis scores from repeated trials for that model.
46 gpt_scores = [
47     -5.25, -5.63, -6, -6.38, -6, -5.25, -5.63, -6.88, -7.5, -6.38,
48     -7, -6.38, -6.5, -6.38, -6.38, -7.75, -7, -7.5, -5.88, -7,
49     -3.84, -3.98, -5, -4.88, -4.88, -4.88, -4.88, -5.26, -4.88, -5.5,
50     -4.88, -4.88, -4.88, -5.64, -5.38, -4.88, -5.38, -5.26, -4.62, -5.64
51 ]
52
53 claude_scores = [
54     -7.38, -5.88, -6.38, -5.88, -5.88, -5.63, -6.13, -5.88, -6.13, -6.97,
55     -6.13, -7.01, -6.24, -6.86, -5.89, -6.76, -7.32, -6.55, -7.12, -6.04,
56     -2.7, -2.56, -4.1, -3.98, -4.62, -3.46, -3.2, -2.94, -3.58, -3.2,
57     -3.08, -3.46, -3.2, -3.84, -3.98, -3.72
58 ]
59
60 deepseek_scores = [
61     -7.13, -6.88, -6.88, -6.5, -6.5, -7.13, -7.13, -7.13, -7.13, -7.75,
62     -6.38, -6.63, -7.5, -7.13, -6.88, -7.75, -6, -7.75, -7.23, -6.89,
63     -5.26, -3.46, -5.9, -3.84, -3.98, -3.58, -6.92, -5, -3.08, -5.76,
64     -5.76, -5.52, -5, -3.2, -3.72, -5, -5.26, -5.52, -2.56, -5.12
65 ]
66
67
68 # -----
69 # 1) One-Way ANOVA
70 # -----
71 # Tests whether the three model groups share the same population mean.
72 f_stat, p_val = f_oneway(gpt_scores, claude_scores, deepseek_scores)
73

```

```

74 print("=== One-Way ANOVA (Economic Scores) ===")
75 print("F-statistic:", round(f_stat, 4))
76 print("P-value:", round(p_val, 4))
77
78 if p_val < 0.05:
79     print("Result: Significant difference between group means (p < 0.05)")
80 else:
81     print("Result: No significant difference between group means (p ≥ 0.05)")
82
83 print("-----")
84
85 # -----
86 # 2) Tukey HSD (Post-Hoc)
87 # -----
88 # Performs pairwise comparisons among models with family-wise error control.
89 all_scores = gpt_scores + claude_scores + deepseek_scores
90 labels = (
91     ["GPT-4"] * len(gpt_scores) +
92     ["Claude"] * len(claude_scores) +
93     ["DeepSeek"] * len(deepseek_scores)
94 )
95
96
97 tukey_result = pairwise_tukeyhsd(
98     endog=np.array(all_scores, dtype=float),
99     groups=np.array(labels, dtype=str),
100     alpha=0.05
101 )
102
103 print("=== Tukey HSD (pairwise comparisons) ===")
104 print(tukey_result)

```

A.2 Social Scores: ANOVA + Tukey HSD

```

1  # -*- coding: utf-8 -*-
2  """Social One Way ANOVA Test + Turkey HSD Test
3
4  Automatically generated by Colab.
5
6  Original file is located at
7      https://colab.research.google.com/drive/1ev0B_3tmTWjsYBFJF6SBHYayK96n0TpE
8  """
9
10 """
11 ANOVA + Tukey HSD on LLM Social Ideology Scores
12
13 Purpose
14 -----
15 This script tests whether three LLMs (GPT-4, Claude, DeepSeek) differ in their
16 mean *social* ideology scores (e.g., libertarian/authoritarian axis after
17 standardization), using:
18
19 1) One-way ANOVA:
20     - Tests the null hypothesis that all group means are equal.
21
22 2) Tukey's HSD (post-hoc):
23     - If ANOVA suggests a significant difference, Tukey HSD identifies *which*

```

```

24     pairs of groups differ while controlling family-wise error rate.
25
26 Context
27 -----
28 In the paper, the inferential pipeline is described as:
29 - One-Way ANOVA -> (if significant) Tukey HSD, run separately for economic and
30   social scores. This file is the "social scores" version. :contentReference[oaicite
      :1]{index=1}
31
32 Dependencies
33 -----
34 - SciPy: f_oneway for ANOVA
35 - statsmodels: pairwise_tukeyhsd for Tukey HSD
36 - NumPy: array handling / formatting
37
38 How to run
39 -----
40 python social_anova_tukey.py
41 """
42
43 from scipy.stats import f_oneway
44 from statsmodels.stats.multicomp import pairwise_tukeyhsd
45 import numpy as np
46
47 # -----
48 # Input Data: Social Scores
49 # -----
50 # Each list contains the social-axis scores from repeated trials of the
51 # political typology tests for that model.
52 #
53 # NOTE: These reflect the same cleaned/standardized dataset described in
54 # the methodology (e.g., after any preprocessing/outlier rules you applied).
55 gpt_scores = [
56     -5.69, -5.49, -5.23, -5.33, -4.56, -4.67, -5.08, -5.54, -5.69, -5.33,
57     -5.18, -6.31, -5.28, -6.46, -6.26, -6.41, -5.95, -6.62, -5.9, -5.95,
58     -4.1, -3.87, -4.25, -4.33, -3.93, -4.45, -4.37, -3.99, -3.35, -4.46,
59     -4.11, -4.06, -3.74, -3.71, -4.47, -4.01, -4.11, -4.39, -3.41, -4.17
60 ]
61
62 claude_scores = [
63     -7.03, -6.82, -6.87, -6.77, -6.26, -6.35, -5.85, -6.77, -6.67, -6.88,
64     -6.15, -5.79, -6.53, -6.94, -5.47, -6.29, -5.96, -6.71, -5.38, -5.63,
65     -2.11, -2.94, -3.35, -3.53, -4.04, -3.13, -2.89, -2.75, -3.21, -3.04,
66     -3.03, -3.25, -3.08, -3.55, -3.63, -3.25
67 ]
68
69 deepseek_scores = [
70     -6.41, -6.1, -6, -5.74, -5.59, -6.62, -6.15, -6.1, -6.21, -6.6,
71     -5.85, -5.85, -6.56, -5.49, -6.41, -6.77, -4.97, -6.15, -5.42, -6.34,
72     -3.38, -3.02, -4.82, -3.19, -3.5, -2.18, -5.21, -3.51, -2.51, -4.66,
73     -4.28, -4.16, -4.08, -3.74, -3.17, -3.74, -3.95, -3.92, -2.24, -3.59
74 ]
75
76
77
78 def run_one_way_anova(*groups: list[float]) -> tuple[float, float]:
79     """
80     Run a one-way ANOVA across >=2 groups.
81

```

```

82 Parameters
83 -----
84 *groups : list[float]
85     Each group is a list of numeric observations.
86
87 Returns
88 -----
89 (f_stat, p_val) : tuple[float, float]
90     F-statistic and p-value from SciPy's f_oneway.
91 """
92 f_stat, p_val = f_oneway(*groups)
93 return float(f_stat), float(p_val)
94
95
96 def run_tukey_hsd(all_scores: list[float], labels: list[str], alpha: float = 0.05):
97     """
98     Run Tukey's Honestly Significant Difference test.
99
100    Parameters
101    -----
102    all_scores : list[float]
103        Flattened observations from all groups.
104    labels : list[str]
105        Group label for each observation (same length as all_scores).
106    alpha : float
107        Significance level (default 0.05).
108
109    Returns
110    -----
111    statsmodels result object (printable table)
112    """
113    return pairwise_tukeyhsd(
114        endog=np.array(all_scores, dtype=float),
115        groups=np.array(labels, dtype=str),
116        alpha=alpha
117    )
118
119
120 def main() -> None:
121     """
122     Execute the ANOVA + Tukey HSD workflow and print results.
123
124     Interpretation logic:
125     - If  $p < 0.05$ , reject null: at least one group mean differs.
126     - Otherwise, fail to reject null: no evidence of mean differences.
127     """
128     # -----
129     # 1) One-way ANOVA
130     # -----
131     f_stat, p_val = run_one_way_anova(gpt_scores, claude_scores, deepseek_scores)
132
133     print("=== One-Way ANOVA (Social Scores) ===")
134     print("F-statistic:", round(f_stat, 4))
135     print("P-value:", round(p_val, 4))
136
137     if p_val < 0.05:
138         print("Result: Significant difference between group means ( $p < 0.05$ )")
139     else:
140         print("Result: No significant difference between group means ( $p \geq 0.05$ )")

```

```

141
142 print("-----")
143
144 # -----
145 # 2) Tukey HSD (post-hoc)
146 # -----
147 # Tukey is often reported after ANOVA to locate which pairs differ.
148 # You can choose to only run Tukey when ANOVA is significant; here we run it
149 # unconditionally for transparency (common in reporting), but you can gate it
150 # behind `if p_val < 0.05:` if you prefer.
151 all_scores = gpt_scores + claude_scores + deepseek_scores
152
153 # Group labels must align 1:1 with observations.
154 labels = (
155     ["GPT-4"] * len(gpt_scores) +
156     ["Claude"] * len(claude_scores) +
157     ["DeepSeek"] * len(deepseek_scores)
158 )
159
160 tukey_result = run_tukey_hsd(all_scores, labels, alpha=0.05)
161
162 print("=== Tukey HSD (pairwise comparisons) ===")
163 print(tukey_result)
164
165
166 if __name__ == "__main__":
167     main()

```